5.3 Model Validation and Verification

Swapnil Sukhdeorao Shinde

3SU2026 AIML-500-01N1B- Machine Learning Fundamentals

Dr. Lawrence Iweriebor

August 9th, 2026

Report: Predictive Modeling for DCC Medication Regimens

Introduction

This report reviews the study **“Predicting an Optimal Medication/Prescription Regimen for Patient Discordant Chronic Comorbidities Using Multi-Output Models”** by Ichchha Pradeep Sharma, Tam V., Shruti Ajay Singh and Tom Ongwere. It evaluates the project’s technical pipeline, ethical safeguards, and alignment with fundamental machine learning practices, offering actionable insights for future deployments.

Technical Rigor: Key Project Steps

Data Preprocessing:

The researchers sourced data via Amazon Mechanical Turk (MTurk) and Qualtrics, capturing clinical deliberation strategies from healthcare providers treating patients with type 2 diabetes and discordant chronic comorbidities (DCCs) like arthritis and depression.

- **Cleaning & Quality Assurance**: Clinical experts vetted MTurk responses to remove nonsensical or irrelevant data. Missing data and outliers were removed, yielding a final clean dataset of 3,931 instances.
- **Feature Engineering & Encoding:** Data was structured using binary inputs to represent the presence or absence of specific diseases and patient concerns (e.g., cost, weight gain, interactions). Categorical and textual variables were transformed using encoders to allow processing by machine learning models.

Model Choice:

The researchers benchmarked four algorithms: Random Forest, K-Nearest Neighbors (KNN), AdaBoost, and XGBoost.

- **Selection:** Random Forest was chosen as the optimal model.
- **Justification:** The authors justified this by highlighting the algorithm’s ensemble nature (using 100 decision trees), which effectively captures complex, non-linear relationships among interacting clinical features while demonstrating robustness against overfitting and noisy data.

Performance Metrics:

The primary metric utilized to evaluate model efficacy was Accuracy.

- **Results:** In the final iteration, Random Forest achieved an accuracy of 93.3%, outperforming KNN (78.5%), XGBoost (76.4%), and AdaBoost (67.3%).
- **Trade-off Analysis:** While accuracy offers interpretability, it is a flawed standalone metric for clinical datasets, which are often imbalanced. The authors acknowledge this limitation, noting that future iterations must include precision, recall, F1-score, and AUC-ROC to provide a more nuanced evaluation of predictive capabilities.

Validation Techniques:

An iterative, three-phase approach was employed to validate and refine the models.

- **Iteration 1 & 2:** Evolved from a standard single-output classifier (which performed poorly) to a multi-class, multi-output classification model capable of predicting a combination of medications simultaneously.
- **Iteration 3 (Optimization):** The researchers utilized a Grid Search technique to test various combinations of hyperparameters (e.g., tree depth, estimators) to locate the optimal configuration for maximum performance.

Current Safeguards & Gaps:

The study primarily contributes to medical ethics by empowering patients. It achieves this by designing systems that prioritize personal concerns like cost and side effects, which ultimately reduces treatment conflicts. To ensure data validity, the research team had local clinical experts rigorously vet all crowdsourced data. However, the paper fails to deeply explore underlying data bias, such as whether MTurk provider demographics skewed the training data. It also overlooks the privacy concerns tied to deploying these models in real-world clinical decision support systems.

Proposed Post-Deployment & Monitoring Plan:

To align with current AI safety standards in healthcare, the following concrete strategies should be implemented post-deployment:

1. **Fairness & Bias Audits:** Conduct routine disparate impact analyses to ensure the model does not recommend sub-optimal (e.g., less effective or higher side-effect) medications to marginalized populations based on latent socioeconomic variables.
2. **Privacy Preservation:** Enforce strict HIPAA compliance via federated learning or differential privacy, ensuring patient input queries cannot be reverse engineered to expose protected health information (PHI).
3. **Continuous Drift Monitoring:** Deploy a monitoring dashboard to track data drift (changes in patient demographics or concerns over time) and concept drift (the introduction of new medications or updated clinical guidelines).

4. **Human-in-the-Loop (HITL) Feedback:** Integrate a clinician override mechanism where rejected AI recommendations are logged and used to retrain the model continuously

Reflection on Machine Learning Fundamentals

Alignment with Recommended Practices:

The authors demonstrated strong alignment with core ML fundamentals in their iterative development cycle. Transitioning from single-output models to multi-class, multi-output wrappers directly addressed the high dimensionality of the target space. Furthermore, their use of correlation matrices for feature analysis and Grid Search for hyperparameter tuning reflects standard best practices for model optimization. Below, we evaluate their approach in the context of our course's recommended frameworks.

- **Foundational Knowledge:** Medication selection for patient comorbidities as a multi-output classification problem. Successfully captured joint probability distributions across multiple target variables instead of isolating targets.
- **Algorithm Exploration:** Benchmarked Random Forest, KNN, AdaBoost, and XGBoost using multi-output wrappers. Random Forest performed best (93.3% accuracy) due to its handling of non-linear boundaries, but the analysis omitted crucial feature-importance metrics.
- **Model Development Lifecycle:** Followed a standard pipeline (baseline, multi-output expansion, Grid Search tuning). However, it suffered from metric myopia (over-relying on accuracy instead of minority class metrics) and weak validation (used simple train/test splits instead of rigorous k-fold cross-validation).
- **Data Challenges:** Filtered crowdsourced data down to 3,931 expert-validated instances. Remaining risks include latent sampling bias from crowdsourcing and information loss from converting complex patient concerns into binary inputs.
- **Real-World Applications:** Demonstrated strong potential for Clinical Decision Support Systems (CDSS). The tool enhances workflows and shared decision-making by tailoring medication combinations to patient-specific priorities.
- **Ethics and Responsibility:** Promoted patient autonomy well. However, clinical safety requires adding HIPAA compliance, demographic fairness audits, and mandatory human-in-the-loop (HITL) clinician oversight.

Insights into Future Work

Insights to Apply:

- **Multi-Output Framing:** The transition to a multi-class, multi-output architecture is highly effective for complex, intersecting problems. I will apply this architecture when modeling scenarios where users have compounding, overlapping constraints (like concurrent diseases).
- **Domain-Expert Data Filtering:** The inclusion of local clinical experts to validate crowdsourced (MTurk) data is a brilliant strategy for utilizing cost-effective data collection without sacrificing domain validity.

Insights to Avoid:

- **Over-reliance on Accuracy:** In future diagnostic or recommender systems, I will avoid using accuracy as the sole guiding metric. Models must be built from the ground up using F1-scores, precision-recall curves, and cost-sensitive loss functions that penalize false negatives (e.g., missing a severe drug interaction).
- **Uncalibrated Confidence:** As the authors noted, many ML models struggle to provide well-calibrated predictions. Future implementations should avoid deploying models that output hard binary recommendations; instead, they should output probability distributions with confidence intervals to aid clinician judgment.

References:

Sharma, I. P., Nguyen, T. V., Singh, S. A., & Ongwere, T. (2024). Predicting an Optimal Medication/Prescription Regimen for Patient Discordant Chronic Comorbidities Using Multi-Output Models. *Information*, *15*(1), 31. https://doi.org/10.3390/info15010031

Wasylewicz ATM, Scheepers-Hoeks AMJW. Clinical Decision Support Systems. 2018 Dec 22. In: Kubben P, Dumontier M, Dekker A, editors. Fundamentals of Clinical Data Science [Internet]. Cham (CH): Springer; 2019. Chapter 11. Available from: https://www.ncbi.nlm.nih.gov/books/NBK543516/ doi: 10.1007/978-3-319-99713-1_11

AI Usage Statement:

Gemini was used to specifically assist with text summarization, paragraph compression, and editorial refinement